



Predator RF

Operator Guide

Pick this up cold. Zero to a working RF picture in under 30 minutes.

This is the single document you need. It assumes nothing — no SDR experience, no Linux, no scripting. If you've used a smartphone and plugged in a USB stick, you can run Predator RF.

1. What this is (in one paragraph)

Predator RF is a phone-first software-defined-radio cockpit. You plug a USB SDR into an Android phone, point an antenna at the world, and the app shows you what's transmitting in real time — frequency, power, location (when GPS is up), and whether you've seen it before. You can mark hits, record a "what's normally here" baseline so the next sweep flags only what's new, and — optionally — link to other phones / Raspberry Pi sensors over a private network to cover a larger area as a small team or solo operator with distributed nodes. There is **no transmit**. The build is RX-only by design.

2. The minimum kit (you need ALL of these to do anything)

Item	Why
Android phone, Android 10 (API 28) or newer	Runs the app
USB-C OTG cable or adapter (USB-C → USB-A female)	Connects the SDR
A USB SDR — RTL-SDR Blog v4 (\$40) is the cheapest, HackRF One (\$340) is the upgrade	The actual radio
An antenna — even the rubber-duck that came with the SDR works	Without one you receive nothing
The Predator RF APK	The app itself — built from the GitHub repo or supplied by your team

That's it. No internet required after install. No subscription. No account.

A full tiered bill of materials (cheap → fleet) is in § 11.

3. First-time install (sideload)

Predator RF is **not on the Play Store**. You sideload the APK.

1. On the phone: **Settings → About phone → tap "Build number" 7 times** until it says "Developer mode is ON".
2. **Settings → Developer options → turn on Install via USB and USB debugging.**
3. Get the APK onto the phone (USB cable from a computer, Google Drive, email — anything).
4. On the phone, open the file (Files app → Downloads → tap app-debug.apk).
5. When Android asks "Install unknown apps?" → **Allow** for whatever app you opened the APK from → **Install**.
6. Open **Predator RF**.

When the app first launches it will ask for these permissions — **grant all of them**:

- **Location** ("While using the app") → for the GPS-stamped hit map
- **Storage / Files** → for saving baselines and exporting CSVs
- **USB device access** → pops up the first time you plug in your SDR; tap **Always allow for this device**

If you skip any of these the app still runs, but the matching feature goes dead-quiet (no map fix, no exports, no SDR).

4. Plug in the radio and start receiving

1. Connect your SDR to the phone with the USB-C OTG adapter. Antenna goes on the SDR.
2. The first plug-in pops a system dialog: **"Open Predator RF when this USB device is connected?"** — check the box and tap **OK**.
3. In the app, on the left rail tap the **source dropdown** (top-left, says None until a radio is selected) → pick RTL-SDR, Airtspy, HackRF, etc. matching what you plugged in.
4. Tap ► **Start Listening** (big button under the source dropdown).
5. The waterfall on the right starts scrolling. **You're live.**

If the waterfall stays black, see § 10 (Troubleshooting).

5. The eight tabs (left-side rail, top to bottom)

The whole app is organized into eight tabs. They're labeled by their 3- or 4-letter code on the rail. Tap to switch.

Code	Name	What it's for
SPEC	Spectrum	Live waterfall + tuner. Where you actually look at signals.
HITS	Hits & Events	Every signal the app has noticed, plus the running event log.
NET	Network	Catalog of known networks / TGs / aliases — your "rolodex of emitters."
MAP	Map	GPS-stamped hits on a map view.

MIS	Mission Config	Mission mode (Manual / Classify / Scan / QuickScan), search bands, targets, excludes, dwell timing.
KUJ	Kujhad Fleet	Link this phone to other Predator RF phones / Raspberry Pi sensors over a network.
SYS	System	App settings — modules, themes, decoder bridges, ATAK/CoT, baseline comparison.
BASE	Baseline	Record what the noise floor / normal traffic looks like, save it, then suppress it next time so only NEW signals fire as hits.

6. The five workflows you'll actually use

6.1 Just look around (passive recon)

1. **SPEC** tab → set the source → ► Start Listening.
2. Drag the waterfall left/right to retune. Pinch to zoom the view bandwidth.
3. Tap a peak to drop a marker on it.
4. To name what you just found: tap the marker → **Assign Marker** → tap "**Tap to edit**" in the popup → type a label.

6.2 Mark a hit and send it to your team

1. With a marker on a signal, tap **Assign Marker** if it isn't already.
2. The marker becomes a **hit** — visible on the **HITS** tab.
3. To send it to ATAK / TAK: **SYS** tab → **ATAK / CoT** section → enter your TAK server IP + port → **Enable CoT**. The hit appears on the ATAK map as a friendly contact at your phone's current GPS location plus a chat message with the freq + power.

6.3 Record a baseline so you only see NEW signals

The first time you set up in a new area, run a baseline so the app learns what's "normally here" (paging towers, nearby commercial radios, broadcast leakage). Then any signal NOT in the baseline is flagged as new.

1. **BASE** tab → **Frequency Ranges** → set Range Name, Start Hz, Stop Hz → + **Add Range**. Repeat for each band you care about. (Tap "**From Current View**" to grab whatever the waterfall is showing right now.)
2. **Recording** section → leave filename blank for auto-naming, OR tap to set a custom one → tap ► **START RECORDING** (green).
3. Let it run for at least 5 minutes (longer = better — 30 min for a thorough sweep).
4. ■ **STOP RECORDING** → **Save to File**.
5. **SYS** tab → **Baseline Comparison** → load the file you just saved → enable **Scan against baseline**. Set a threshold (default 6 dB). Now only signals exceeding the baseline by that margin become hits.

6.4 Run an automated scan across multiple bands

1. **MIS** tab → **Mission Mode** → **Scan**.
2. **Search Bands** section → add the bands you want swept (Name + Start Hz + Stop Hz). Tap **+ Add Band**.
3. **Targets** section → optional — frequencies you specifically want to flag as priority.
4. **Excludes** section → optional — frequencies to ignore (your own gear, known broadcast carriers).
5. **Dwell + QuickScan Delay + Duration** — defaults are sane. Increase dwell if you're chasing weak intermittent signals.
6. Back to **SPEC** → ► **Start Listening**. The app will now sweep your bands automatically and drop hits on the **HITS** tab as it finds them.

6.5 Link a second phone or RPi sensor

This is the Kujhad Fleet feature — distributed sensors with one operator screen.

Make this phone discoverable:

1. **KUJ** tab → **Listen** section → set Listen port (default 8080), Device name, API key (use a long random string — anything works, just make it match on both sides).
2. Toggle **Listen** on. The phone now serves its state to peers.

Add a peer (another phone or RPi running the same protocol):

1. **KUJ** tab → **Add Peer** → Name (anything memorable), Host (the peer's IP), Port (their listen port), API Key (must match theirs), tap **Add peer**.
2. The peer appears in the peer list. Toggle the **Mirror peer spectrum** switch to view their waterfall on your screen while still controlling your local SDR.

For the full Kujhad architecture (TLS, ZeroTier overlays, RPi node setup) see docs/OPERATOR_RUNBOOK.md — it covers the Linux / Python backend side of this same fleet.

7. Editing fields on a touchscreen

Every editable field (peer host, port, frequency, range name, target, filename, etc.) works the same way:

1. Tap the field. A popup opens at the **top of the screen**, above the keyboard.
2. The keyboard slides up automatically.
3. Type. You can see what you're typing — the popup is intentionally pinned high so the keyboard never covers it.
4. Tap **OK** (or hit Enter) to commit. **Cancel** to discard.

If a field doesn't open a popup when tapped, that's a bug — restart the app once and try again. If it persists, file it.

8. The orange / yellow / red status bar at the top

Color	Meaning
Green / dim grey	Nominal
Orange	Thermal warning — the SoC is heating up; expect frame drops
Red	Severe — back off (lower sample rate, close other apps, get the phone out of direct sun)

The phone will throttle itself before it cooks, but reduce load proactively when you see orange.

9. Field-day checklist (print this)

Before you leave:

- [] Phone charged + power bank packed
- [] APK version verified (open app → SYS → check version against your team's current build)
- [] SDR + antenna + USB-C OTG cable in the bag
- [] At least one **baseline file** for the area you're going to (record it ahead if possible — see § 6.3)
- [] If using ATAK: TAK server reachable + credentials entered + a successful test beacon yesterday
- [] If using Kujhad fleet: peers added, all on the same overlay network (ZeroTier / Tailscale / LAN), API keys match

On site, bring-up:

1. Antenna up first — give it sky if you can.
2. Plug SDR into phone → confirm USB permission dialog accepts.
3. Open Predator RF → SPEC → pick source → ► Start.
4. Verify waterfall is moving + GPS lock indicator is green (top status bar).
5. **MIS** → set mission mode for the day's job.
6. If using baseline: **SYS** → Baseline Comparison → load file → enable.
7. If using Kujhad: **KUJ** → toggle Listen → confirm each peer shows green/connected.
8. If using ATAK: **SYS** → ATAK/CoT → enable. Verify your icon appears on the TAK map within 30 s.

During mission (every 30 min):

- Glance at the thermal indicator
- Glance at GPS lock
- Glance at hit count on **HITS** — climbing = system working
- Phone battery > 20% — swap to power bank before you hit 10%

End of mission:

1. **MIS** → mode back to Manual.
2. **HITS** → export hits to CSV (button at the top of the tab).
3. **BASE** → if you recorded a fresh baseline, save it before powering down.

4. ► Stop Listening → close app → unplug SDR.

10. Troubleshooting

Symptom	Most likely cause	Fix
Waterfall is black after Start Listening	SDR not selected as source, or USB permission was denied	Source dropdown → re-pick. Unplug + replug SDR → tap "Always allow" on the dialog.
App says "device busy"	Another app (or a previous Predator RF session) has the SDR	Force-stop Predator RF in Settings → Apps; unplug + replug
Touch feels unresponsive / wrong widget fires	Old build — this was fixed in the touch-passthrough + ID-stack patches	Update the APK to the current build
Soft keyboard covers the input field	Old build — fixed by IME-inset clamp	Update APK
GPS never locks	Phone needs sky view; indoor / urban canyon = no fix	Step outside; wait 60 s
Kujhad peer shows red / disconnected	Network not reachable, or API keys don't match	Ping the peer from a terminal app; confirm both phones have the same API key character-for-character
ATAK marker never appears	Wrong server IP/port, or the TAK server is filtering by UID prefix	Check TAK server logs; verify CoT enabled and not in dry-run
Phone hot, framerate drops	Thermal throttling	Get out of the sun; reduce sample rate (SPEC → source settings)
App crashes on startup after install	APK was sideloaded over a different signing key	Uninstall the old version completely (Settings → Apps → Predator RF → Uninstall), then reinstall

11. Bill of materials — pick your tier

Prices are USD, May 2026, ballpark.

Tier 0 — Bare minimum (~\$50)

You probably already own most of this.

Item	Approx	Notes
Android phone you already own	\$0	Anything Android 10+
RTL-SDR Blog v4	\$40	Cheapest viable SDR. 500 kHz - 1.75 GHz.
USB-C OTG adapter	\$8	USB-C male → USB-A female
Rubber-duck antenna	included	Comes with the v4
Total	~\$48	

What you can do: receive everything from FM broadcast through ~1.7 GHz, mark hits, record baselines, run scans. No HF, no transmitting. No fleet.

Tier 1 — Solo field operator (~\$300-450)

Adds wider coverage, better front-end, and external GPS so the map works indoors-ish.

Item	Approx	Notes
Tier 0 kit	\$48	
Airspy Mini or HackRF One	\$130 / \$340	Mini = better dynamic range up to 1.7 GHz; HackRF = 1 MHz - 6 GHz, the workhorse
Telescoping whip antenna (Diamond RH-77CA)	\$50	Vastly better than rubber-duck
External USB GPS (BU-353N5 or similar)	\$35	Works through the same OTG hub; faster lock than phone GPS
Powered USB-C OTG hub	\$25	So the phone doesn't drain its battery powering the SDR
20,000 mAh USB-C PD power bank	\$40	A full mission day
Total	~\$330 (Airspy) / ~\$540 (HackRF)	

What you can do: the above plus HF (HackRF), full ATAK integration, all-day battery life, GPS that locks fast.

Tier 2 — Solo TOC with one remote sensor (~\$1,400-1,800)

Adds a Raspberry Pi sensor node running the Path 1 Python backend so you can drop a sensor somewhere and walk the perimeter while it collects.

Item	Approx	Notes
Tier 1 kit (HackRF flavor)	\$540	
Raspberry Pi 5 8GB + case + active cooler + PSU	\$130	The sensor node
256 GB A2 microSD	\$25	
Second RTL-SDR Blog v4 (for the Pi)	\$40	
GPS HAT (Adafruit Ultimate GPS HAT, or USB GPS)	\$45	Sensor node needs its own fix
Linux laptop (anything you already own)	\$0-\$500	Operator workstation runs the same Python backend OR you skip it and run from the phone
ZeroTier or Tailscale (overlay network)	\$0	Free tier covers a small fleet
Outdoor antenna mount + 25 ft LMR-400 + N-to-SMA pigtails	\$80	Get the antenna off the truck
Discone or wideband vertical (Diamond D130J)	\$130	The actual perimeter antenna
Pelican 1450 case	\$130	So the Pi survives the field
Total	~\$1,400 + laptop	

What you can do: above plus distributed collection, long-duration unattended overwatch, mission ledger, post-mission CSV export & after-action.

Tier 3 — Small team / multi-node fleet (~\$5,000+)

Repeat Tier 2's RPi sensor as many times as you have positions to cover, plus higher-end antennas + comms gear.

Item	Approx	Notes
Tier 2 kit	\$1,400	
3× additional RPi sensor nodes (each tier-2 RPi+SDR+GPS+case)	\$1,500	
Cellular hotspot (Netgear M6 Pro) + month of data	\$300	If LAN/overlay won't cover the area
Or: small mesh radio link (goTenna Pro X2, RAJANT, etc.)	\$500-\$5,000	When cell is denied
Commercial-grade band-specific antennas (VHF dipole, UHF Yagi for direction-finding)	\$400	When you know what band you live in
Tactical mast (10-25 ft fiberglass, military surplus)	\$150	Get the antenna into the air
Ruggedized Pelican / Storm cases × N	\$400	One per node
Operator laptop (Lenovo ThinkPad / Dell rugged)	\$500-\$2,000	If your phone-only setup isn't enough
Total	~\$5,000-\$10,000	Scales with fleet size

What you can do: above plus full-perimeter overwatch from a single operator screen, ATAK integration to higher-echelon TOC, after-action ledger across all nodes, and a credible answer to "where else is that emitter being heard from."

12. Where to go next

- **docs/OPERATOR_RUNBOOK.md** — the Linux Python backend (Path 1) deep-dive. Read this when you set up RPi sensor nodes or the operator workstation.
- **docs/MISSION_READY_CHECKLIST.md** — the longer pre-mission checklist with the Python-side commands. Print before a real op.
- **docs/SIDELoad_README.md** — step-by-step APK install from a Windows box.

- **docs/ATAK_COT_FORMAT.md** — the exact CoT XML this app emits (for TAK plugin authors).
- **docs/1_conops.md** — the doctrine document. Read this if you're deciding whether Predator RF fits a problem at all.

If something in this guide is wrong or out of date, fix it and commit. This document is the contract with the operator.